THE ASSOCIATION OF MATHEMATICS TEACHERS OF INDIA

58th NMTC - SCREENING TEST - BHASKARA CONTEST

NMTC - JUNIOR LEVEL - IX & X GRADES

PART – A

Note: Only one of the choices A, B, C, D is correct for each question. Shade the alphabet of your choice in the response sheet. If you have any doubt in the method of answering, seek the guidance of the supervisor.

1. If $x \neq y \neq z$ with $x + y + z = 0$ and $x^3 + x + 1 = 0$, $y^3 + y + 1 = 0$, $z^3 + z + 1 = 0$, then the numerical value of $x^9 + y^9 + z^9$ is

(A) 7 (B) –2 (C) – 6 (D) 6

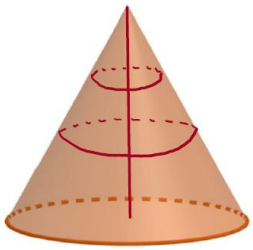
2. The number of natural numbers n satisfying the equation

$$\frac{8^n - 2^n}{6^n - 3^n} = 2, \text{ is}$$

(A) only one (B) two

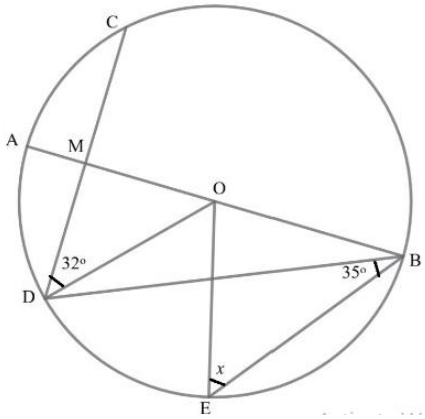
(C) four (D) infinitely many

3. A right circular cone is cut into three parts of volume v_1, v_2, v_3 from the vertex respectively, by giving two cuts parallel to the base, which trisect the altitude of the cone. Then $v_1 : v_2 : v_3$ =



(A) 1:2:3 (B) 1:4:9 (C) 1:8:27 (D) 1:7:19

4. In the adjoining figure, AOB is a diameter of the circle with center O. Point M is the mid-point of chord CD. ∠CDO = 32º, ∠DBE = 35 º and ∠BEO = x º, then x =

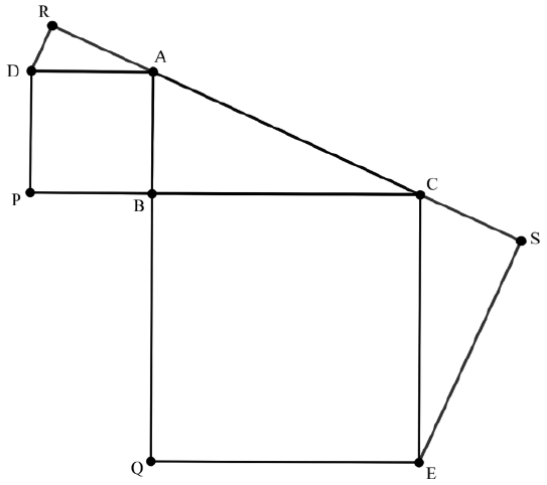


(A) 74º (B) 54º (C) 64º (D) 66º

5. During a wartime, a group of n soldiers had enough food to last for 30 days. After 10 days, 50 more soldiers joined. If the food now can only last for 16 days, the value of n is

(A) 180 (B) 200 (C) 220 (D) 310

6. Let ABC be a right-angled triangle, ∠ABC = 90°. AB = 5cm, BC = 12cm. Squares ABPD and BCEQ are drawn in the exterior of the ΔABC. DR is drawn perpendicular to line AC at R and ES is perpendicular to the line AC at S. Let DR = h_1, and ES = h_2, then $\frac{h_1 + h_2}{AC}$ =



(A) $\frac{5}{13}$ (B) $\frac{12}{13}$ (C) 2 (D) 1

7. If $(\sqrt{2})^{\sqrt{x}} - (\sqrt{2})^{\sqrt{y}} = 48$, where x and y are natural numbers, then the value of x + y is

(A) 208 (B) 112 (C) 132 (D) 96

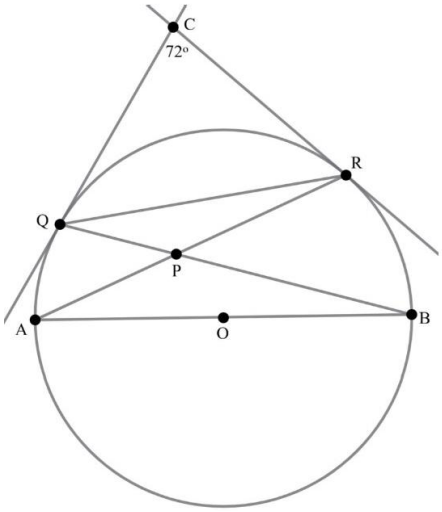
8. x and y are real numbers which satisfy $x^2 + y^2 + 2x + 6y + 10 = 0$. Then the numerical value of $\frac{x+y}{x-y}$ is

(A) 1 (B) 2 (C) –2 (D) $\frac{-1}{2}$

9. If $x = \frac{\sqrt{101}+1}{2}$, then the find the value of $(x^3 - 26x - 23)^5$ is

(A) 1 (B) $\frac{1}{32}$ (C) 32 (D) 81

10. In the adjoining figure, AOB is a diameter of the circle with center O. QR is a chord and the tangents to the circle at points Q and R include an angle of measure 72°. Chords BQ and AR intersect at P, them the measure of ∠RPB is



(A) 40° (B) 34° (C) 38° (D) 36°

11. If the arithmetic mean of two numbers is 25 and their geometric mean is 24, then an equation with the given numbers as its roots, is

(A) $x^2 + 50x + 576 = 0$ (B) $x^2 - 50x - 576 = 0$

(C) $x^2 + 50x - 576 = 0$ (D) $x^2 - 50x + 576 = 0$

12. The number of real values of x satisfying the equation $\frac{3x^2 - 21x}{x^2 - 7x} = x - 4$, is

(A) Zero (B) One (C) Two (D) Three

13. The average marks obtained by Sunil in 6 subjects is 88. On subsequent verification, it was found that the marks obtained by him in a subject was wrongly copied was 86 instead of 68. The correct average marks obtained by him is $(ab)_{10}$, that is a two-digit number in base 10. Then $a^2 + b^2 =$

(A) 89 (B) 88 (C) 100 (D) 113

14. The graph of $x^2 + y^2 = 25$ is plotted in a real xy-plane. The number of points (x, y) on this graph such that both x and y are integers, is

(A) 8 (B) 12 (C) 16 (D) 10

15. If $(n + 1) = \frac{2 \times f(n) + 1}{2}$, for all natural numbers n and $f(1) = \frac{1}{2}$,

then $-f(1) + f(2) - f(3) + f(4) + f(5) + \ldots\ldots - f(99) + f(100) =$

(A) 75 (B) 50 (C) – 25 (D) 25

PART – B

Note: Write the correct answer in the space provided in the response sheet.

16. In ΔABC, $\angle ACB = 30^\circ$, AB = 5cm. Then the circumradius of ΔABC is _____ cm.

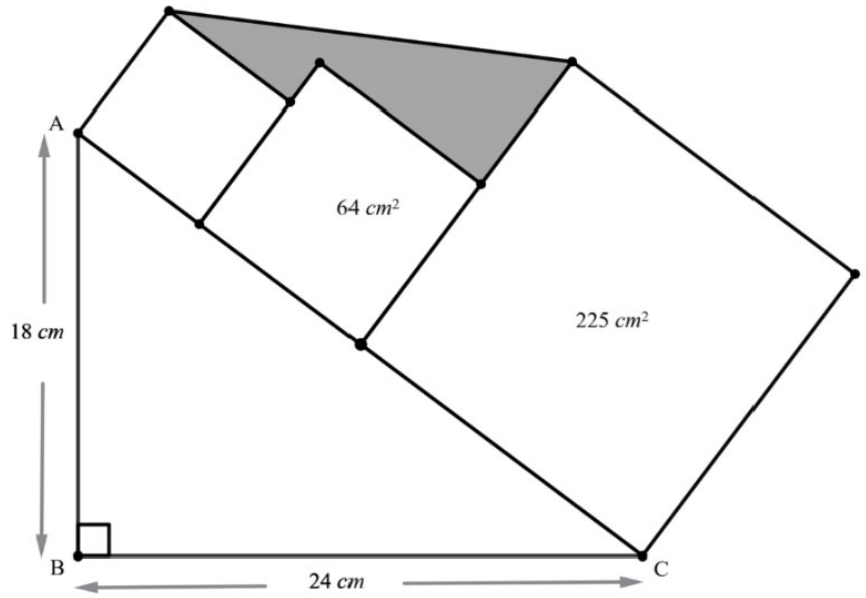
17. If $\frac{\sqrt{60} + \sqrt{95} + \sqrt{228} + 12}{\sqrt{5} + 2\sqrt{12} + \sqrt{19}} = \frac{\sqrt{a} + \sqrt{b}}{c}$

where a, b are square-free natural numbers and c is a natural number, then the value of $a + b + 7c =$ _____.

18. Let $\frac{3x - 25}{x^2 - 5x + 6} = \frac{a}{x - 3} + \frac{b}{x - 2}$

be an identity on x. Then the numerical value of $b - a$ is _____.

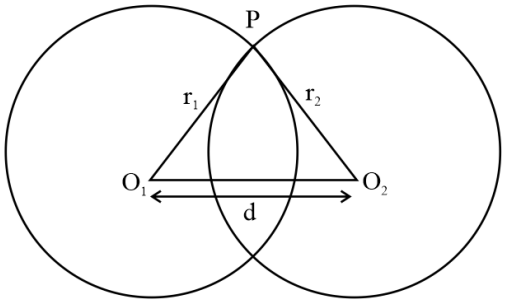
19. In the adjoining figure, ABC is a right angled triangle. ∠ABC = 90°, AB = 18cm, BC = 24cm. There are three squares drawn externally on the hypotenuse AC as shown, such that AC would be the sum of the side lengths of these squares. Area of two of these squares is 225cm^2, and 64cm^2 respectively, as indicated in the figure. Then the area of the shaded region = _____ cm^2.



20. We say that a positive integer is arithmetically sequenced if its digits, in order, form an arithmetic sequence. Then the number of 4-digit positive integers which are arithmetically sequenced, is _____.

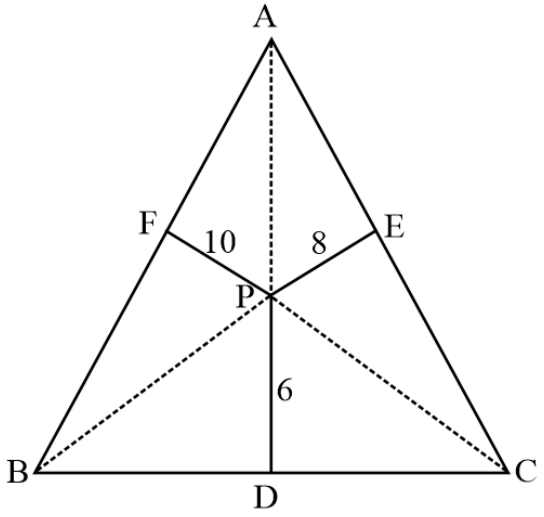
21. The smallest positive integer $(abc)_{10}$, in base 10, a, b, c are digits, with $500 > (abc)_{10} > 150$ such that the number formed by all possible arrangements (for example $(bac)_{10}$, $(cba)_{10}$) of all the digits of $(abc)_{10}$, is prime, is _____.

22. Let circle-1 and circle-2 are orthogonal circles. The length of an arc measuring 60° of circle-1 is equal to the length of an arc measuring 45° of circle-2. Then the ratio of the area of square on the line-segment joining their centres to the area of square on the radius of circle-1, in its simplest form, is _____.
[Note: Let two distinct circles are intersecting in different points. If their tangents at the point of intersection are perpendicular to each other, then the circles are said to be Orthogonal circles]



23. If m and n are the roots of the equation $x^2 + 2mx + 5n = 0$; $m \neq 0$, $n \neq 0$, then the sum of roots of this equation is _____.

24. P is a point in the interior of an equilateral ΔABC. Line-segments PD, PE, PF are the perpendiculars drawn from P to the sides BC, CA, AB respectively with D lies on side BC, E lies on side CA and F lies on side AB. If PD = 6cm, PE = 8cm, PF = 10cm, then area of the triangle ABC is _____ cm^2.



25. Let the product 12 × 15 × 16 = 3146 in base b, where b is a real number. Then the sum s = 12 + 15 + 16 in base b is _____.

26. If $\frac{4^x}{2^{x+y}} = 8, \frac{9^{x+y}}{3^y} = 243$ then $x^2 + xy + 7y^2$ = _____.

27. Let $a_1, a_2, a_3, \ldots, a_n$ be all natural numbers and also $n \in \mathbb{N}$ such that $a_1 + a_2 + a_3 + \ldots + a_n = 1000$. When $a_1 \times a_2 \times a_3 \times \ldots \times a_n$ would be maximum, then n = _____.

28. If x = 2026, and y = $\frac{1}{1013}$, then the value of $\left(\frac{x^2+1}{y}\right)\left(\frac{y^2+1}{x}\right)+\left(\frac{x^2-1}{y}\right)\left(\frac{y^2-1}{x}\right)$ is _____.

29. If $\frac{1}{3^2+1}+\frac{1}{4^2+2}+\frac{1}{5^2+3}+\ldots\ldots=\frac{a}{b}$
where a and b are coprime natural numbers, then a + b = _____.

30. n is a natural number such that (n + 17) and (n – 40) both are perfect square numbers, then the square root of greatest possible value of (n – 40) is _____.